Solution End Sem Exam MCC505 (Prob + Stat)

1(a)

$n = 100, \quad \bar{x} = 8, \quad \sigma = \sqrt{10.5}, \quad \sigma^2 = 10.5$

$n_1 = 50, \quad \bar{x}_1 = 10, \quad \sigma_1 = 2$

$n_2 = 50, \quad \bar{x}_2 = ?, \quad \sigma_2 = ?$

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{(n_1+n_2)} \Rightarrow \bar{x}_2 = \frac{(n_1+n_2)\bar{x} - n_1\bar{x}_1}{n_2}$$

$$\Rightarrow \bar{x}_2 = \frac{100\times 8 - 50\times 10}{50} = \frac{800-500}{50} = \frac{300}{50} = 6$$

$$\boxed{\bar{x}_2 = 6} \text{ —— (1 mark)}$$

$d_1 = \bar{x}_1 - \bar{x} = 10 - 8 = 2, \quad d_2 = \bar{x}_2 - \bar{x} = 6 - 8 = -2$

combined variance $$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1+n_2)}$$

$$\Rightarrow 10.5 = \frac{50(4+4) + 50(\sigma_2^2+4)}{100}$$

$$\Rightarrow 1050 = 400 + 50\sigma_2^2 + 200$$

$$\Rightarrow \sigma_2^2 = \frac{1050-600}{50} = \frac{450}{50} = 9$$

$$\Rightarrow \boxed{\sigma_2 = 3} \text{ —— (2 marks)}$$

1(b)

Karl Pearson's coefficient of skewness $S_K = \frac{\text{Mean} - \text{Mode}}{SD}$

$\Rightarrow S_K = \frac{\bar{x} - M_o}{\sigma}$, Given $\bar{x} = 120, \quad M_o = 123$

$S_K = -0.3$

Hence $-0.3 = \frac{120-123}{\sigma}$

$$\Rightarrow \sigma = \frac{-3}{-0.3} = 10 \text{ —— } (\tfrac{1}{2} \text{ mark})$$

$$\therefore CV = \frac{SD}{\text{Mean}} = \frac{10}{120} = 0.083 \text{ —— } (\tfrac{1}{2} \text{ mark})$$

$$\boxed{CV = 0.083} \xrightarrow{\frac{1}{2}+\frac{1}{2}} \text{(1 mark) (Total)}$$

1(c)

$x \mid f$, $N = \sum_{i=1}^{n} f_i$

$\mu_4(x) = \frac{1}{N}\sum_{i=1}^{n} f_i (x_i - \bar{x})^4$. Let $U = \frac{X-A}{h}$, A + h are s.c. constants.

which gives $X = A + hU$, $\bar{X} = A + h\bar{U}$

$\Rightarrow$ $X - \bar{X} = h(U - \bar{U})$

$\therefore$ $\mu_4(x) = \frac{1}{N}\sum_{i=1}^{n} f_i h^4 (u_i - \bar{u})^4 = h^4 \cdot \frac{1}{N}\sum_{i=1}^{n} f_i (u_i - \bar{u})^4$

$\Rightarrow$ $\boxed{\mu_4(x) = h^4 \cdot \mu_4(u)}$, depends upon scale only.

(2 marks)

1(d)

using the transformation $U = \frac{X-32}{2}$, we get

$N = \sum f_U = 400$, $\sum fu = -91$, $\sum fU^2 = 2435$, $\sum fU^3 = -1915$

$\sum fU^4 = 32339$

Hence the moments about origin of the variable U is obtained as

$\mu_1'(u) = \frac{1}{N}\sum fu = -0.2275$, $\mu_2'(u) = \frac{1}{N}\sum fU^2 = 6.0875$

$\mu_3'(u) = \frac{1}{N}\sum fU^3 = -4.7875$, $\mu_4'(u) = \frac{1}{N}\sum fU^4 = 80.8475$

The moments about mean (central moments) of the variable U are

$\mu_1(u) = 0$, $\mu_2(u) = \mu_2'(u) - \mu_1'^2(u) = 6.0357$

$\mu_3(u) = \mu_3'(u) - 3\mu_2'(u)\mu_1'(u) + 2\mu_1'^3(u) = -0.6563$

$\mu_4(u) = \mu_4'(u) - 4\mu_3'(u)\mu_1'(u) + 6\mu_2'(u)\mu_1'^2(u) - 3\mu_1'^4(u) = 78.3732$

— ($\frac{1}{2}$ mark)

Hence, $\mu_1(x) = 0$,

Here, $h = 2$ $\mu_2(x) = h^2 \mu_2(u) = 24.1428$ — (1 mark)

$\mu_3(x) = h^3 \mu_3(u) = -5.2504$ — (1 mark)

$\mu_4(x) = h^4 \mu_4(u) = 1253.9712$ — ($1\frac{1}{2}$ mark)

$\beta_1(x) = \frac{\mu_3^2(x)}{\mu_2^3(x)} = 0.001959$, $\mu_3(x) < 0$ -vly skewed ($\frac{1}{2} + \frac{1}{2} = 1$ mark)

$\beta_2(x) = \frac{\mu_4(x)}{\mu_2^2(x)} = 2.1514$ < 3 Platykurtic

($\frac{1}{2} + \frac{1}{2} = 1$ mark)

2.(a) Let D be the event that the person has the disease and 'T' be the event that the berson is diagnosed as having the diseas[e]

We need to compute the conditional probability $P(D|T)$.

Using Bayes Theorem,

Given $P(D)=0.1$
$\therefore P(D^c)=0.9$
Also, $P(T|D)$
$=1-0.01$
$=0.99$

$$P(D|T)=\frac{P(T|D)\,P(D)}{P(T|D)P(D)+P(T|D^c)\,P(D^c)}$$

$$=\frac{0.99\times0.1}{0.99\times0.1+0.05\times0.9}$$

$$=0.69.$$

So given that the person has tested positive the probability that this person actually has the disease is $\boxed{0.69}$. $\longrightarrow$ 2M

(b) The distribution function of the RV X is $F(x)=P(X\le x)$, $x\in\mathbb{R}$
$$=\int_{-\infty}^{x} f(t)\,dt$$

$$\therefore F(x)=\begin{cases}0, & \text{if } x\le 0\\ \int_0^x t\,dt=\frac{x^2}{2} & \text{if } 0<x\le 1\\ \int_0^1 t\,dt+\int_1^x(2-t)\,dt=2x-\frac{x^2}{2}-1 & \text{if } 1<x\le 2\\ 1 & \text{if } x\ge 2\end{cases}$$ $\longrightarrow$ 1M

Then $P(0.3\le X\le 1.5)=P\{X\le 1.5\}-P\{X<0.3\}$
$$=P\{X\le 1.5\}-P\{X\le 0.3\}$$
$$=F(1.5)-F(0.3)$$
$$=\boxed{0.83}$$ $\longrightarrow$ 1M

$$E(X)=\int_{-\infty}^{\infty}x f(x)\,dx$$
$$=\int_0^1 x.x\,dx+\int_1^2 x.(2-x)\,dx$$
$$=\left[\frac{x^3}{3}\right]_0^1+\left[x^2-\frac{x^3}{3}\right]_1^2$$
$$=\frac{1}{3}+3-\frac{7}{3}=\boxed{1}$$ $\longrightarrow$ $\frac{1}{2}$M

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$= \int_0^1 x^2 \cdot x\,dx + \int_1^2 x^2 (2-x)\,dx$$

$$= \left[\frac{x^4}{4}\right]_0^1 + \left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_1^2$$

$$= \frac{1}{4} + \frac{2}{3}\times 7 - \frac{1}{4}\times 15$$

$$= \frac{14}{3} - \frac{14}{4}$$

$$= \frac{7}{6}$$

$$\therefore V(X) = E(X^2) - \{E(X)\}^2$$

$$= \frac{7}{6} - 1$$

$$\therefore \boxed{V(X) = \frac{1}{6}} \longrightarrow \frac{1}{2}M$$

(c) The individual probability density functions of X_1 and X_2 are given by

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2)\,dx_2$$

$$= \int_0^{\infty} x_1 x_2 . \exp\left[-\frac{1}{2}(x_1^2 + x_2^2)\right] dx_2$$

$$= x_1 e^{-\frac{1}{2}x_1^2} \cdot \int_0^{\infty} x_2 e^{-\frac{1}{2}x_2^2}\,dx_2$$

$$= x_1 e^{-\frac{1}{2}x_1^2} \cdot \int_0^{\infty} e^{-\frac{1}{2}x_2^2}\,d\left(\frac{x_2^2}{2}\right)$$

$$= x_1 e^{-\frac{1}{2}x_1^2} \cdot \left[-e^{-\frac{1}{2}x_2^2}\right]_0^{\infty}$$

$$\therefore \boxed{f_{X_1}(x_1) = x_1 e^{-\frac{1}{2}x_1^2}, \quad 0 < x_1 < \infty} \longrightarrow \frac{1}{2}M$$

Similarly, $f_{X_2}(x_2) = x_2 e^{-\frac{1}{2}x_2^2}$, $0 < x_2 < \infty$ $\longrightarrow \frac{1}{2}M$

The probability that each sound intensity is less than or equal to 1 is given by

$$P[X_1 \le 1, X_2 \le 1] = \int_{-\infty}^{1}\int_{-\infty}^{1} f(x_1, x_2)\, dx_1\, dx_2$$

$$= \int_0^1\int_0^1 x_1 x_2 e^{-\frac{1}{2}(x_1^2+x_2^2)}\, dx_1\, dx_2$$

$$= \left(\int_0^1 x_1 e^{-\frac{1}{2}x_1^2} dx_1\right)\left(\int_0^1 x_2 e^{-\frac{1}{2}x_2^2} dx_2\right)$$

$$= \left(1-e^{-\frac{1}{2}}\right)\left(1-e^{-\frac{1}{2}}\right) = \left(1-e^{-\frac{1}{2}}\right)^2 = \boxed{0.1548} \longrightarrow 1\tfrac{1}{2}M$$

$$\because \int_0^1 x_1 e^{-\frac{1}{2}x_1^2} dx_1 = \int_0^1 e^{-\frac{1}{2}x_1^2} d\left(\frac{x_1^2}{2}\right) = \left[-e^{-\frac{x_1^2}{2}}\right]_0^1 = 1-e^{-\frac{1}{2}}$$

The probability that the sum of the sound intensities is less than 1 is given by

$$P[X_1+X_2 \le 1] = \iint_{\{(x_1,x_2):\, x_1+x_2\le 1\}} f(x_1, x_2)\, dx_1\, dx_2$$

$$= \int_{x_1=0}^{1}\int_{x_2=0}^{1-x_1} x_1 x_2 e^{-\frac{1}{2}(x_1^2+x_2^2)}\, dx_2\, dx_1$$

$$= \int_0^1 x_1 e^{-\frac{x_1^2}{2}}\left(\int_{x_2=0}^{1-x_1} x_2 e^{-\frac{x_2^2}{2}} dx_2\right) dx_1$$

$$= \int_0^1 x_1 e^{-\frac{x_1^2}{2}}\left(\int_0^{1-x_1} e^{-\frac{x_2^2}{2}} d\left(\frac{x_2^2}{2}\right)\right) dx_1$$

$$= \int_0^1 x_1 e^{-\frac{x_1^2}{2}}\left(\left[-e^{-\frac{x_2^2}{2}}\right]_0^{1-x_1}\right) dx_1$$

$$= \int_0^1 x_1 e^{-\frac{x_1^2}{2}}\left[1-e^{-\frac{(1-x_1)^2}{2}}\right] dx_1$$

$$= \int_0^1 x_1 e^{-\frac{x_1^2}{2}} dx_1 - \int_0^1 x_1 e^{-\frac{1}{2}\{x_1^2+(1-x_1)^2\}} dx_1$$

$$= 1-e^{-\frac{1}{2}} - \int_0^1 x_1 e^{-x_1^2+x_1-\frac{1}{2}} dx_1$$

$$= \boxed{0.034} \longrightarrow 1\tfrac{1}{2}M$$

(d) Let $X_i = \begin{cases} 1, & \text{if ith toss is head} \\ 0, & \text{otherwise.} \end{cases}$

Since the X_i's are i.i.d. Bernoulli distributed,

$$\mu = EX_i = \frac{1}{2}, \quad \sigma^2 = Var(X_i) = p(1-p) = \frac{1}{2}\left(1-\frac{1}{2}\right) = \frac{1}{4}.$$

Now let $S_{400} = X_1 + X_2 + \cdots + X_{400}$

$= \sum_{i=1}^{400} X_i \Rightarrow$ count the number of heads we get out of 400 coin tosses

$\therefore E(S_{400}) = 400\mu = 400 \times \frac{1}{2} = 200$

$V(S_{400}) = 400\sigma^2 = 400 \times \frac{1}{4} = 100$

By the Central Limit Theorem, $\dfrac{S_{400} - 200}{\sqrt{400\sigma^2}} \sim N(0,1)$

1½M

$$\therefore P(S_{400} \leq 190) = P(S_{400} - 400\mu \leq 190 - 400 \times \tfrac{1}{2})$$

$$= P\left(\frac{S_{400} - 200}{\sqrt{400\sigma^2}} \leq \frac{190 - 200}{10}\right)$$

$$= P(Z \leq -1)$$

$$= \Phi(-1)$$

$$= 1 - \Phi(1)$$

$$= 1 - 0.8413$$

$$= \boxed{0.1587}$$

1½M

3.

a) Additive property of Gamma distribution:
The sum of independent Gamma variates is also a Gamma variate, i.e., if $X_1, X_2, \dots\dots, X_k$ are independent Gamma variates with parameters $\lambda_1, \lambda_2, \dots\dots, \lambda_k$ respectively then $X_1 + X_2 + \cdots + X_k$ is also a Gamma variate with parameter $\lambda_1 + \lambda_2 + \cdots + \lambda_k$ [1]

b) Let $X = X_1 + X_2$. By the additive property of Poisson distribution X is also a Poisson variate with parameter (say) $\lambda = 2 + 6 = 8$ [1]
Hence the probability of x calls in-between 10 A.M. and 12 noon is given by
$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-8}8^x}{x!}$; $x = 0,1,2, \dots$
Probability that more than 5 calls come in between 10 A.M. and 12 noon is given by
$P(X > 5) = 1 - P(X \leq 5) = 1 - \sum_{x=0}^{5} \frac{e^{-8}8^x}{x!} = 1 - 0.1912 = 0.8088$ [1]

c)

1st part:
Moment generating function of Normal distribution,

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\{t(\mu + \sigma z)\} \exp(-z^2/2)\, dz \qquad \text{where } z = \frac{x-\mu}{\sigma}$$

$$= e^{\mu t} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(z^2 - 2t\sigma z)}{2}\right\} dz$$

$$= e^{\mu t} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(z-\sigma t)^2 - \sigma^2 t^2}{2}\right\} dz$$

$$= e^{\mu t + \sigma^2 t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(z-\sigma t)^2}{2}\right\} dz$$

$$= e^{\mu t + \sigma^2 t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{u^2}{2}\right\} du$$

$$= e^{\mu t + \sigma^2 t^2/2} \frac{1}{\sqrt{2\pi}} \times 2 \times \int_{0}^{\infty} \exp\left\{-\frac{u^2}{2}\right\} du$$

$$= e^{\mu t + \sigma^2 t^2/2} \frac{1}{\sqrt{2\pi}} \times \sqrt{2\pi}$$

$$= e^{\mu t + \sigma^2 t^2/2}$$

Hence $M_X(t) = e^{\mu t + \sigma^2 t^2/2}$ [3]

2nd part:
Given pdf of exponential distribution is $f(x) = e^{-x}$, $x \geq 0$

cdf of the exponential distribution is given by

$$F(x) = \int_0^x f(t)\,dt$$

$$= \int_0^x e^{-t}\,dt$$

$$= [-e^{-t}]_0^x$$

$= 1 - e^{-x}$ [1]

Then the cdf of $X_{(1)}$ and $X_{(n)}$ are given by

$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n = 1 - [1 - (1 - e^{-x})]^n = 1 - e^{-nx}$ [0.5]

$F_{X_{(n)}}(x) = [F(x)]^n = [1 - e^{-x}]^n$ [0.5]

d)

1st part:

Let X be the variable represent the marks obtained by a number of students for a certain subject.

Given that $X \sim N(\mu, \sigma)$ where $\mu = 65$ and $\sigma = 5$

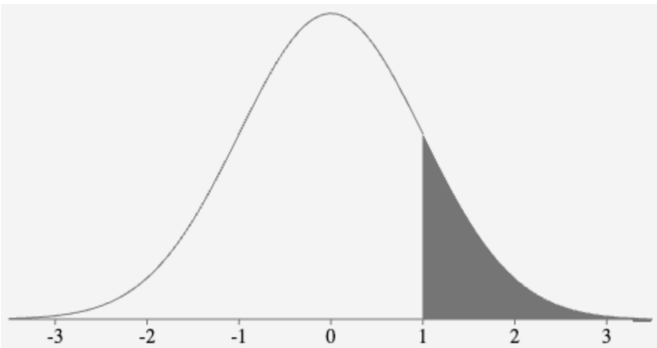
Let $z = \frac{x-\mu}{\sigma} = \frac{x-65}{5}$ then $Z \sim N(0,1)$

Now, $P(X > 70) = P(Z > 1)$

$= 0.5 - P(0 \le z \le 1)$

$= 0.5 - 0.3413$

$= 0.1587$ [1]



Then the probability of exactly 2 of them have marks over 70 is

$3_{C_2}(0.1587)^2(0.8413)^{3-2} = 0.06357$ [1]

2nd part:

Given that a continuous random variable X have a pdf

$$f(x) = \frac{1}{2^{n/2}\,\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-x/2}, \qquad x > 0$$

Now $M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx}\, f(x)\, dx$

$$= \frac{1}{2^{n/2}\,\Gamma(\mathrm{n}/2)} \int_0^\infty e^{tx}\, x^{\frac{n}{2}-1} e^{-x/2}\, dx$$

$$= \frac{1}{2^{n/2}\,\Gamma(\mathrm{n}/2)} \int_0^\infty e^{-\frac{(1-2t)x}{2}}\, x^{\frac{n}{2}-1}\, dx$$

$$= \frac{1}{2^{\frac{n}{2}}\,\Gamma\left(\frac{\mathrm{n}}{2}\right)} \times \left(\frac{1-2t}{2}\right)^{-n/2} \int_0^\infty e^{-u}\, u^{\frac{n}{2}-1}\, du \qquad \text{where } \left(\frac{1-2t}{2}\right) x = u$$

$$= \frac{1}{2^{n/2}\,\Gamma(\mathrm{n}/2)} \times \left(\frac{1-2t}{2}\right)^{-n/2} \times \Gamma(\mathrm{n}/2)$$

$$= (1-2t)^{-n/2} \qquad \text{where } |2t| < 1 \qquad [2]$$

Ans4 (a): $f(x,y) = \frac{1}{3}(x+y), \quad 0 \le x \le 1, \quad 0 \le y \le 2$

Marginal pdf of x:

$$f(x) = \int_0^2 f(x,y)\,dy = \frac{1}{3}\int_0^2 (x+y)\,dy = \left(\frac{xy}{3} + \frac{1}{3}\cdot\frac{y^2}{2}\right)\Big|_0^2$$

$$= \frac{2}{3}(1+x), \quad 0 \le x \le 1$$

marginal pdf of y:

$$f(y) = \int_0^1 \frac{1}{3}(x+y)\,dx = \frac{1}{3}\left(\frac{1}{2}+y\right), \quad 0 \le y \le 2.$$

$$f(y|x) = \frac{f(x,y)}{f(x)} = \frac{1}{2}\left(\frac{x+y}{1+x}\right)$$

$$f(x|y) = \frac{f(x,y)}{f(y)} = \frac{2(x+y)}{1+2y}$$

(i) Two Regression functions:

(i) $$E(Y/x) = \int_0^2 y \cdot f(y|x)\,dy$$

$$= \frac{1}{2(1+x)}\int_0^2 y\cdot(x+y)\,dy$$

$$= \frac{1}{2(1+x)}\left(\frac{xy^2}{2} + \frac{y^3}{3}\right)\Big|_0^2$$

$$E(y|x) = \frac{1}{2(1+x)}\left(2x + \frac{8}{3}\right) = \frac{3x+4}{3(x+1)} \quad -2$$

(ii) $$E(X|y) = \int_0^1 x\, f(x|y)\,dx = \frac{2}{1+2y}\int_0^1 (x^2+xy)\,dx$$

$$= \frac{2+3y}{3(1+2y)} \quad -2$$

$$E(X) = \int_0^1 x\, f(x)\,dx = \frac{5}{9}$$

$$E(X^2) = \int_0^1 x^2 f(x)\,dx = \frac{7}{18}$$

$$Var(X) = \frac{7}{18} - \left(\frac{5}{9}\right)^2 = \frac{13}{162} = \sigma_X^2$$

$= 0.0802$

$$E(Y) = \frac{11}{9}, \quad E(Y^2) = \frac{16}{9},$$

$$Var(Y) = \frac{16}{9} - \left(\frac{11}{9}\right)^2 = \frac{23}{81} = \sigma_Y^2$$

$= 0.2839$

$$E(XY) = \int_0^1 \int_0^2 xy\, f(x,y)\,dx\,dy$$

$$= \frac{1}{3}\left[\left(\int_0^1 x^2\,dx\right)\left(\int_0^2 y\,dy\right) + \left(\int_0^1 x\,dx\right)\left(\int_0^2 y^2\,dy\right)\right]$$

$$= \frac{1}{3}\left[\frac{1}{3} \times 2 + \frac{1}{2} \times \frac{8}{3}\right] = \frac{2}{3}$$

$$Cov(X,Y) = E(XY) - E(X)E(Y) = \frac{2}{3} - \frac{5}{9} \times \frac{11}{9} = -\frac{1}{81}$$

(ii) Two lines of Regression:

1

$$Y - E(Y) = \frac{Cov(X,Y)}{\sigma_X^2}\left(X - E(X)\right) \Rightarrow \boxed{Y - \frac{11}{9} = -\frac{2}{13}\left(X - \frac{5}{9}\right)}$$

$$X - E(X) = \frac{Cov(X,Y)}{\sigma_Y^2}\left(Y - E(Y)\right) \Rightarrow \boxed{X - \frac{5}{9} = -\frac{1}{23}\left(Y - \frac{11}{9}\right)}$$

1

(iii) $r(X,Y) = ?$

$$r(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{-1/81}{\sqrt{\frac{13}{162} \times \frac{23}{81}}} = -\left(\frac{2}{299}\right)^{1/2}$$

1

$= -0.08179$

Ans 4 (b):

$$r_{12} = 0.77, \quad r_{13} = 0.72, \quad r_{23} = 0.52$$

$$r_{12\cdot3} = \frac{r_{12} - r_{13}\, r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}} = \frac{0.77 - 0.72 \times 0.52}{\sqrt{(1-0.72^2)(1-0.52^2)}}$$

$$= 0.\cancel{62} \qquad 0.6674 \qquad 1$$

$$R_{1\cdot23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2 r_{12}\, r_{13}\, r_{23}}{1 - r_{23}^2}$$

$$= \frac{0.77^2 + 0.72^2 - 2 \times 0.77 \times 0.72 \times 0.52}{1 - 0.52^2}$$

$$= 0.7\cancel{3}34 \qquad 0.7329$$

$$R_{1\cdot23} = +0.8\cancel{5}64 \qquad 0.8561 \qquad 1$$

Ans 4 (c): Regression eqn of X_1 on X_2 & X_3 is given by:

$$(X_1 - \bar{X}_1)\frac{w_{11}}{\sigma_1} + (X_2 - \bar{X}_2)\frac{w_{12}}{\sigma_2} + (X_3 - \bar{X}_3)\frac{w_{13}}{\sigma_3} = 0$$

where
$$w = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}$$

$$w_{11} = \begin{vmatrix} 1 & r_{23} \\ r_{23} & 1 \end{vmatrix} = 1 - r_{23}^2 = 1 - (-0.56)^2 = 0.686$$

$$w_{12} = -\begin{vmatrix} r_{21} & r_{23} \\ r_{31} & 1 \end{vmatrix} = r_{13}\, r_{23} - r_{21} = -0.576$$

$$w_{13} = r_{23} r_{12} - r_{13} = -0.56 \times 0.80 - (-0.40) = -0.048$$

Required eqn:

$$\boxed{\frac{0.686}{4.42}(X_1 - 28.02) - \frac{0.576}{1.10}(X_2 - 4.91) - \frac{0.048}{85}(X_3 - 594) = 0}$$

3